Tingyi He

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Education

University of Amsterdam, Anton Pannekoek Institute, Netherlands **2025 – expected July 2027**
MSc Physics and Astronomy, Astronomy and Astrophysics track (joint degree) *Current GPA: 8.27/10.0*

Chongqing University, China **2021 – 2025**
BSc Physics *GPA: 3.35/4.00; ranked 14/157*

University of California, Berkeley, United States **Aug. 2024 – Jan. 2025**
Berkeley Physics International Education Program (BPIE)

Research Experience

Relativistic simulations of standing shocks in the SS 433 jets **Sept. 2026 – expected July 2027**
MSc thesis (in progress) *Formal supervisor: Prof. Oliver Porth*

- Day-to-day advisers: Laura Olivera-Nieto and Joana Kramer.
- Developing relativistic BHAC simulations to determine why standing shocks form near the observed re-brightening regions of the SS 433 jets.
- Testing the roles of disk-wind termination shocks, the W50 ambient medium, and magnetic fields in shaping jet–environment interactions.

Polarimetry and burst energetics of FRB 20240114A **June 2026 – present**
MSc research project (ongoing; manuscript in preparation) *Supervisor: Prof. Jason Hessels*

- Purpose: to analyze whether linear-polarization fraction, polarization-angle behaviour, and rotation measure differ between lower- and higher-energy bursts from FRB 20240114A, while controlling for calibration
- Reproduced the polarization-calibration procedure for PSR J1935+1616, following Kirsten et al. (2021), using PSRCHIVE.
- Applying calibrated polarimetric analysis to FRB 20240114A to relate burst-to-burst polarization and rotation-measure behaviour to its magneto-ionic environment and burst energetics.

Polarization-calibration diagnostics for FRB 20240114A **Oct. 2025 – Mar. 2026**
MSc Astrovaria course research project (completed; grade: 9.0/10.0) *Supervisor: Prof. Jason Hessels*

- Purpose: to determine whether apparent rotation-measure evolution in FRB 20240114A is astrophysical or introduced by polarization-calibration systematics.
- Analysed archival FRB 121102 data with QU fitting and RM synthesis, and assessed the effects of instrumental delay and leakage on rotation-measure estimates.
- Result: showed that instrumental differential delay can mimic rotation-measure shifts.

Numerical simulation of magnetically driven winds from thin discs **Feb. 2025 – June 2025**
Bachelor thesis (completed; grade: 95/100) *Supervisor: Prof. Xiaohong Yang*

- Purpose: to quantify how initial magnetic-field strength affects mass loading, acceleration, and energy and momentum transport in thin-disc winds.
- Constructed and compared two two-dimensional axisymmetric MHD disk-wind models with different initial magnetic-field strengths using PLUTO.
- Computed time-averaged mass-outflow rates and kinetic, thermal, and momentum fluxes to assess the effect of magnetic fields on wind launching and propagation.
- Developed Python post-processing scripts to convert PLUTO binary outputs and analyse large sets of simulation snapshots.

XeBrA: Xenon Breakdown Apparatus **Oct. 2024 – Jan. 2025**
Research assistant, BPIE exchange semester, UC Berkeley *Supervisors: Dr. Ibles Olcina, Prof. Daniel McKinsey*

- Analysed light-sensor data for detector-status monitoring and wrote software to compare SiPM/PMT signals with environmental data, including pressure.
- Refined a peak-finding algorithm for SiPM photon-rate studies and assisted with detector commissioning for a new data-taking campaign.

XENONnT dark-matter experiment **June 2024 – Aug. 2024**
Undergraduate Research Summer Camp project, CUHK-Shenzhen *Supervisor: Prof. Jingqiang Ye*

- Studied the relation between photoionization intensity and the S1 area-fraction-top observable.
- Developed a higher-resolution correction approach based on this correlation.

Publication and Research Outputs

RELICS Collaboration. *Reactor neutrino liquid xenon coherent elastic scattering experiment. Physical Review D* **110**, 072011 (2024). doi:10.1103/PhysRevD.110.072011

Other research outputs: Astrovaria report on FRB 20240114A polarization-calibration diagnostics; PLUTO disk-wind simulation project report; MSc thesis in progress (expected July 2027).

Selected Presentations

- *Rotation-Measure Evolution and Polarization-Calibration Diagnostics for FRB 20240114A*, oral presentation, AstroFlash group meeting, Anton Pannekoek Institute, Mar. 2026.
- *Candidate of Mysterious Dark Matter: Axion-Like Particles*, poster presentation, Physics Directed Reading Report, UC Berkeley, Dec. 2024.
- *Solar ^{8}B Neutrino Search with the XENONnT Dark-Matter Experiment*, oral presentation, undergraduate seminar, UC Berkeley, Oct. 2024.
- *Photoionization Study in the XENONnT Dark-Matter Experiment*, oral presentation, Undergraduate Research Summer Camp, Chinese University of Hong Kong, Shenzhen, Aug. 2024.

Technical Skills

Programming and analysis: Python, C, MATLAB; scientific data analysis, visualization, and simulation post-processing.

Research software: PSRCHIVE, PLUTO, GEANT4; developing familiarity with BHAC.

Research workflows: Linux-based computing, radio-polarization calibration, rotation-measure analysis, scientific writing and presentations.

Languages: Mandarin Chinese (native); English (professional working proficiency).

Honours and Awards

- Science and Technology Academic Innovation Advanced Individual, Chongqing University (Nov. 2022); selected from 169 students in the Department of Physics.
- Second Prize, 7th Southwest College Student Physics Academic Competition (June 2023).
- Outstanding Student, Chongqing University (Oct. 2023).
- Second Prize, 6th Chongqing College Student Physics Innovative Competition (Dec. 2023).
- First Prize, Undergraduate Research Summer Camp, Chinese University of Hong Kong, Shenzhen (Aug. 2024).
- Comprehensive Scholarship for Outstanding Students, Chongqing University: Third Prize (Fall 2021–22; Spring 2022–23) and Second Prize (Fall 2022–23).